

SCENARIO 3

Team 16 Final Report

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Abstract

A peristaltic pump is a type of pump which uses the principle of peristalsis to pump fluids through a flexible tube. In the human body, peristalsis is the involuntary contraction and relaxation of muscles and canals in the digestive system (such as the intestine); which allow fluids and contents to be pushed forward through them in a wave-like movement.¹ Creating peristaltic pumps allows us to use the principle of peristalsis in other external applications, such as processing chemicals, abrasive slurries (in mining), food and beverages, and more.²

The peristaltic pump functions on the basis of rotating rollers (or “shoes”) which compress a tube between them and a backing case. This creates a vacuum which draws the fluid into the tube, leading to a positive displacement action which prevents fluid backflow.

The peristaltic pump has several advantages:

1. They prevent the fluid from flowing backwards, eliminating the need of building check-valves for when the pump is not running³.
2. Only the tube is in contact with the fluid, reducing risks of the pump contaminating the fluid or vice versa. It can therefore transport clean/sterile or aggressive fluids without external exposure.
3. Very low maintenance due to their very simple structure (no valves, seals, and glands), in turn making them cheap to build.
4. They are able to work for several types of fluids (slurries, viscous, shear-sensitive and aggressive fluids) and can be used for different functions (transfer, dose, or metre different fluids).

Peristaltic pumps are crucial in biomedical engineering. They are used to:

1. Pump IV fluids through an infusion device.
2. Maintain blood circulation during coronary artery surgeries: the pumps can be used in heart-lung machines.
3. Hemodialysis systems⁴.

¹ Oxford Languages and Google. Available at: <https://languages.oup.com/google-dictionary-en/>

² Albin Pump. *How Do Peristaltic Pumps Work?*. Available at: <https://www.albinpump.com/en-us/news/how-peristaltic-pumps-work>

³ Watson Marlow Fluid Technology Solutions. *Peristaltic pumps - how do they work?*. Available at: <https://www.wmfts.com/en/support/pump-principles/how-do-peristaltic-pumps-work/#:~:text=Rollers%20or%20shoes%20in%20a,the%20fluid%20contaminating%20the%20pump.>

⁴ Wikipedia. (2023). *Peristaltic pump*. Available at: https://en.wikipedia.org/wiki/Peristaltic_pump

Introduction

The aim of this project is to design and prototype a manually powered (Allen key) peristaltic pump which maximises the volume of liquid pumped over time in a closed circuit. The pump must be made of a breadboard, an allen key crank, a silicone tube, up to 8 bearings and 2 liquid bag containers; and must be smaller or up to the size of the bread board.

The pump's circuit must be closed, and pumping fluid from bag 1 to bag 2, where bag 1 is 5 cm below the pump and bag 2 is 5 cm above it. The pump must rotate for 60 revolutions or 2 minutes, whichever happens first; then the amount of liquid in bag 2 will be measured.

Tolerance Characterization

When designing the pump, the first and most important element that needed to be taken into consideration was the tolerance measurements, which allows adequate choosing the size of the components which make up the peristaltic pump. This ensures they correctly fit together and avoid error margins between what is designed on the CAD software and what is being physically printed in real life. The data used in this section can be found in the appendix.

The 3D Printing Machine

The 3D printer which was used for this project was the Ultimaker 3. The team was provided with four 3D printed “example” shapes, which were made up of a mixture of extrusions (columns), circular holes of different diameters, hexagonal holes of different sizes, and T-holes. Measurements were taken for all elements of the shape, including the shape's lengths and widths in a variety of locations.

To maximise the number of data points per element, each element in the shape was measured four times, once by every operator of the team. This method was done to ensure that an error was not carried forward for all measurements by one operator, and to avoid errors related to the positioning of the calliper. For a better accuracy of tolerance measurements, the team contacted the other groups working on the printer, and incorporated their dataset to ours.

After obtaining all the data points, error margins were calculated for every element by deducing the value given on the technical drawings from the measurements that were physically measured. Then, using excel, the average errors and standard deviations were calculated for every measured feature. The standard deviation of the error told us how precise the printed model was and the average error gave us the accuracy of the final product.

The next step was to identify the ability to combine measurements for the same element/feature to identify patterns.

After some analysis, it was deduced that the final 3D printed design deviated from the desired product produced on fusion 360, potentially due to uncontrollable conditions in the 3D printing machine.

One of the main observations for deviation was that the majority of extrusions on the testing piece had a negative error, meaning that most of the extrusions came out shorter than the intended height when printed. This was due to the gravitational forces acting downwards and pulling the extrusion down in the printing machine, which cannot be controlled and highlighted the tolerance for the height of extrusions for the shafts holding our bearings. These deviations could also be due to incorrect temperature settings and shrinkage of the material as it cools.

Modulus of error vs actual diameter for extrusions

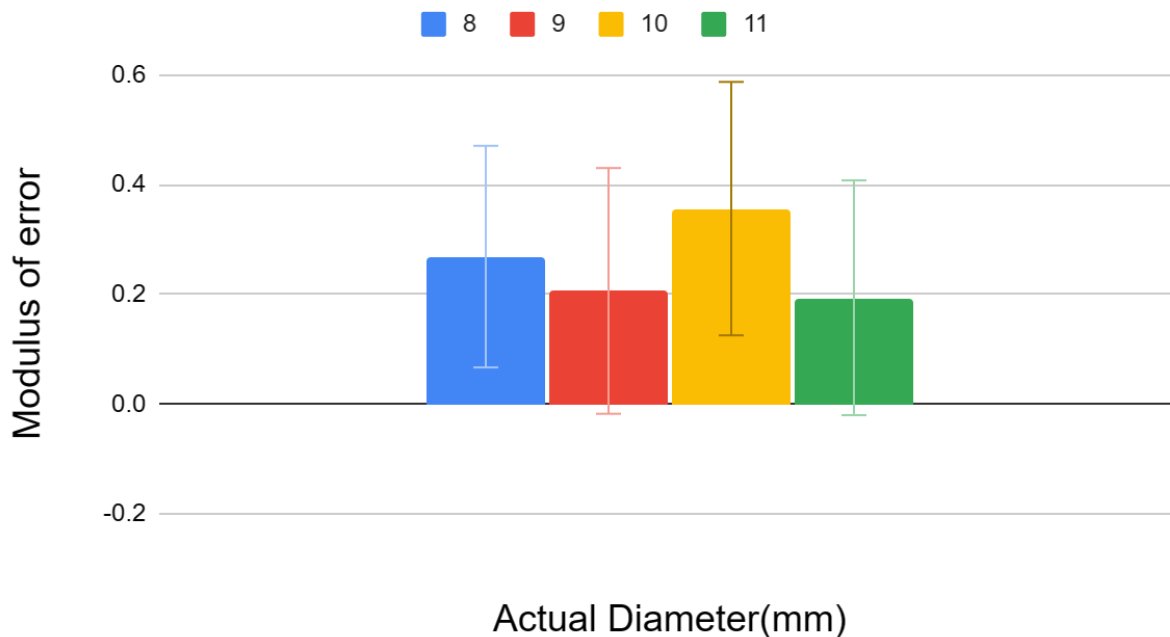


Figure 1. Diagram representing the modulus of the error with respect to the physical diameter of the extrusion

In Figure 1, every colour and number represents a different measured extrusion diameter. The figure shows that there is no correlation between the accuracy and the diameter of the extrusion. It is due to random error and conditions that the 3D printer was set to.

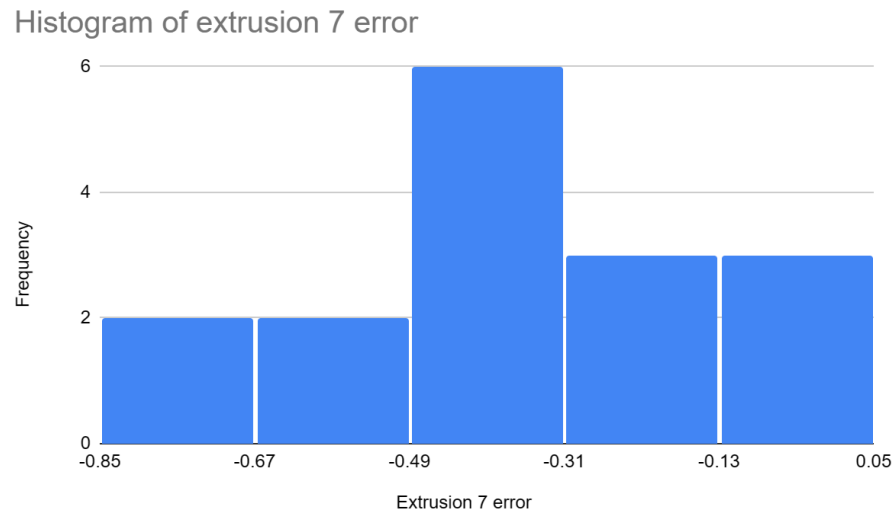


Figure 2. Histogram of the errors recorded showing the standard deviation of errors for extrusion 7.

In Figure 2, the precision is 0.23 and accuracy is -0.36. Additionally, the holes printed were smaller.

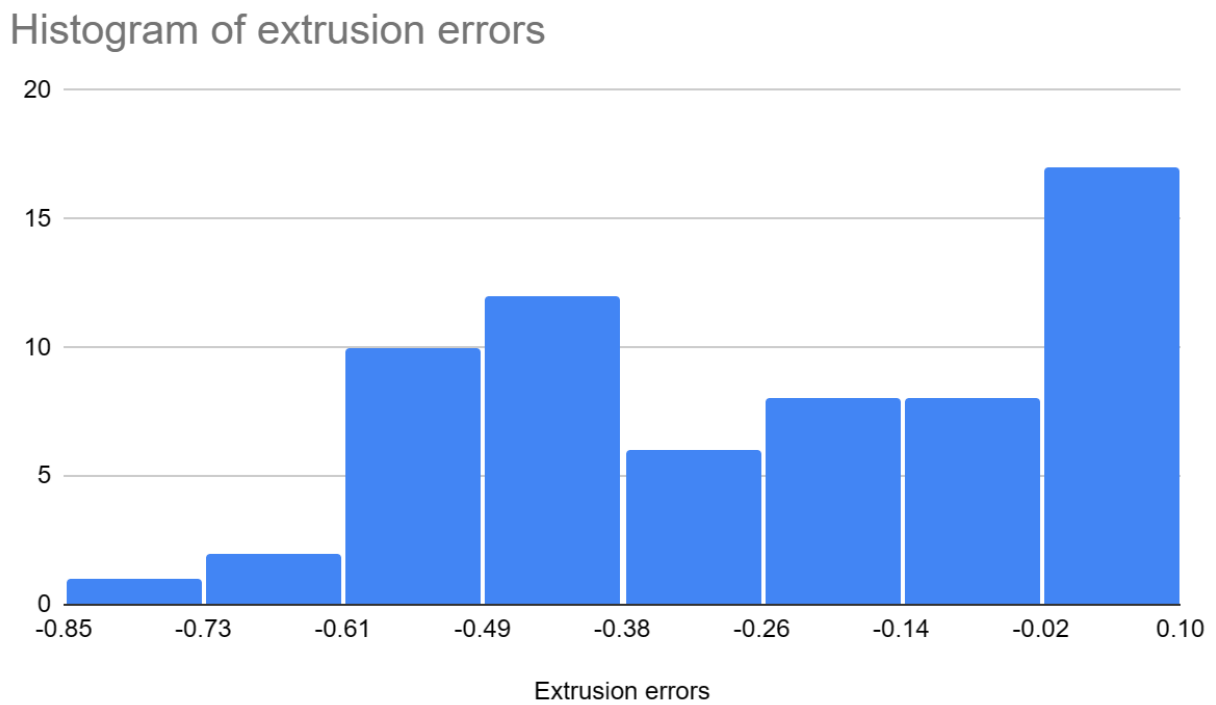


Figure 3. Histogram for all errors of extrusions

Standard deviation vs hole diameter

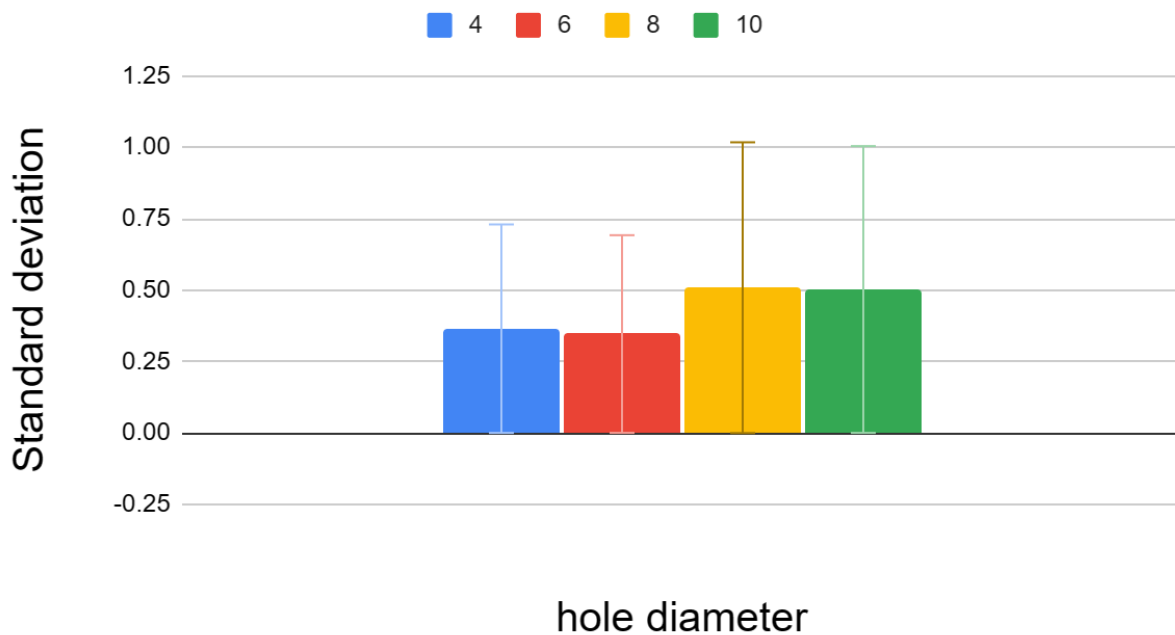


Figure 3. Graph representing the standard deviation of circular holes increasing as the actual diameter increases.

Figure 3 shows that as the intended diameter increases the standard deviation increases. This could also be due to the expansion and warping of the material as it cooled down. This can be prevented by using materials that are less prone to wearing and allow more time instead of immediately cooling it.

With this data we can form a line of best fit and interpolate the standard deviations for the diameter of circles we do not have data for.

Histogram of hole errors

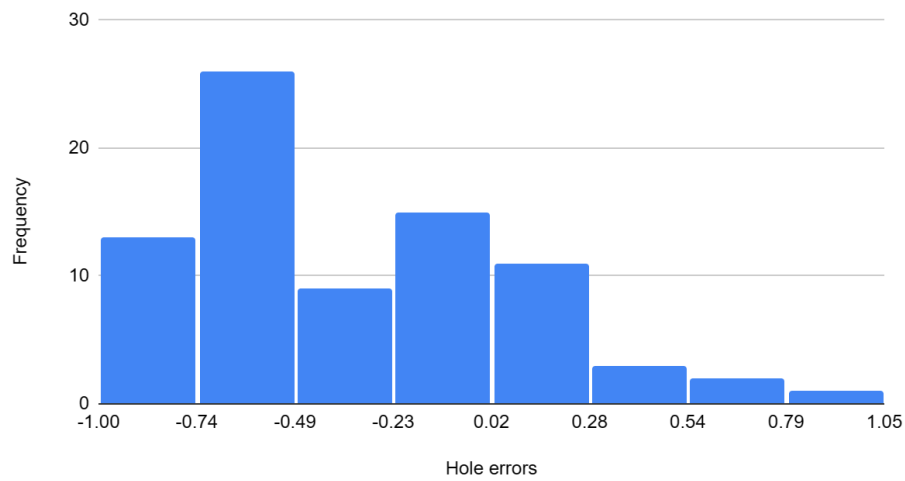


Figure 4. Histogram representing the frequency of errors for holes

The reasons for a larger deviation of errors could be due to how printing a perfect sphere is hard and it will not be perfectly round.

Similarly, figure 5 shows hexagon holes also showed decreases in size after printing.

Histogram of Hexgaons errors

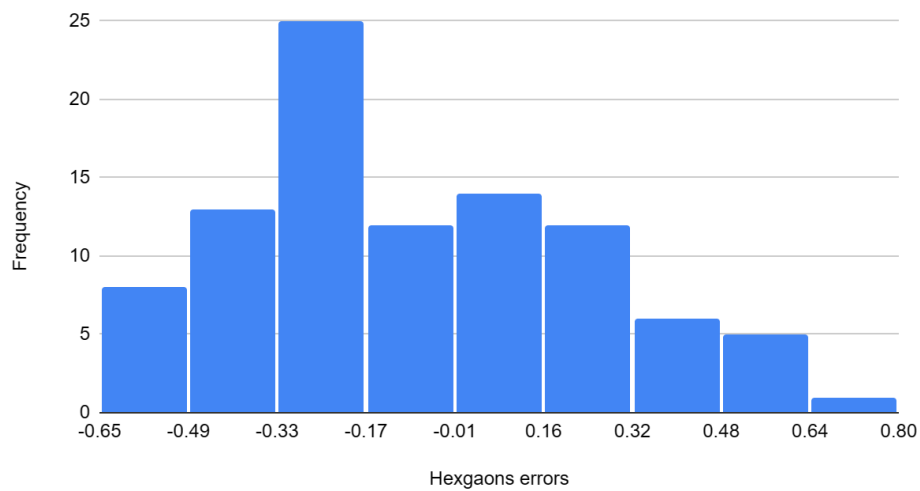


Figure 5. Histogram of all the measurements of the hexagon holes

Modulus of mean error vs hexagon size

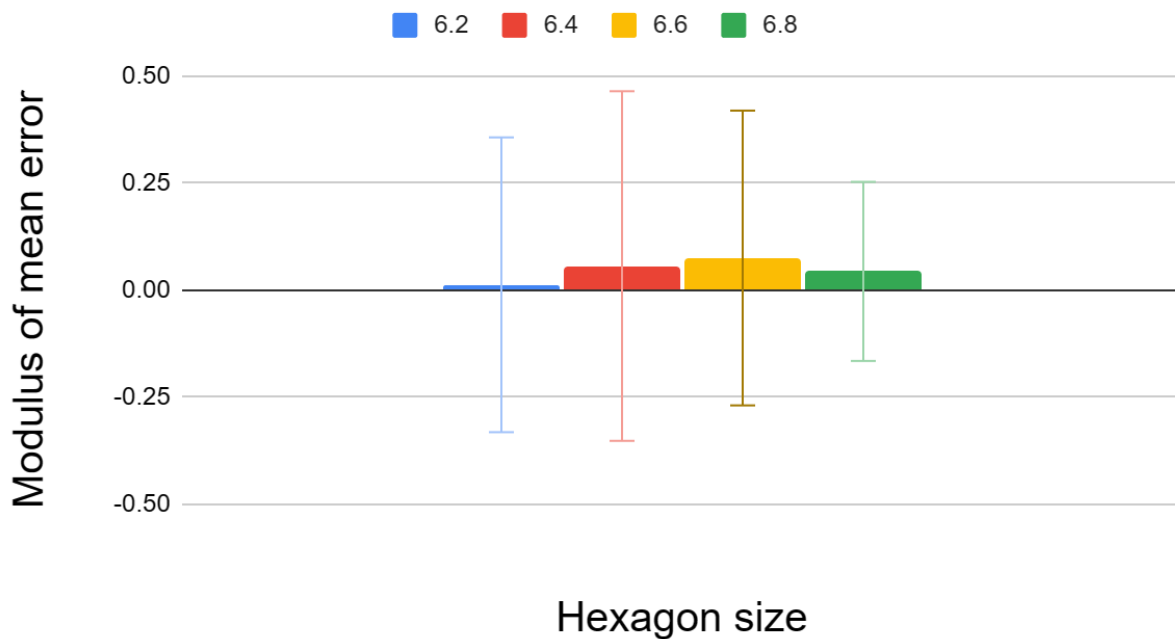


Figure 6 represents the relationship between hexagon size and standard deviation of the size.

Since hexagons have more edges it has to be more finely printed, which can be more challenging, explaining the larger standard deviations compared to the other features.

The table below shows the correlation between the average error and standard deviation, and the feature specified.

Feature	Average error	Standard deviation
Extrusion	No correlation	No correlation
Hexagon	Weak positive correlation	Weak positive correlation
Holes	Strong positive correlation	Strong positive correlation

The mitigating actions taken for the final design was using a transference fit for the extrusions and shafts holding the bearing in place, because the extrusions were smaller than the design specifications. The hexagons had a clearance fit of 0.5 mm. The holes for the caps had a clearance fit of 0.5 mm. The extrusions had a tolerance of 0.5 mm.

The Laser Cutter Machine

The same process used for the 3D printing tolerance measurement was used for the laser cutter machine. The data for this can also be found in the appendix.

The data has proved the laser cutting machine to be more accurate in producing the desired cut out compared to 3D printing as it has lower standard deviations resulting in a high precision and high accuracy.

Mean error vs Diameter (mm)

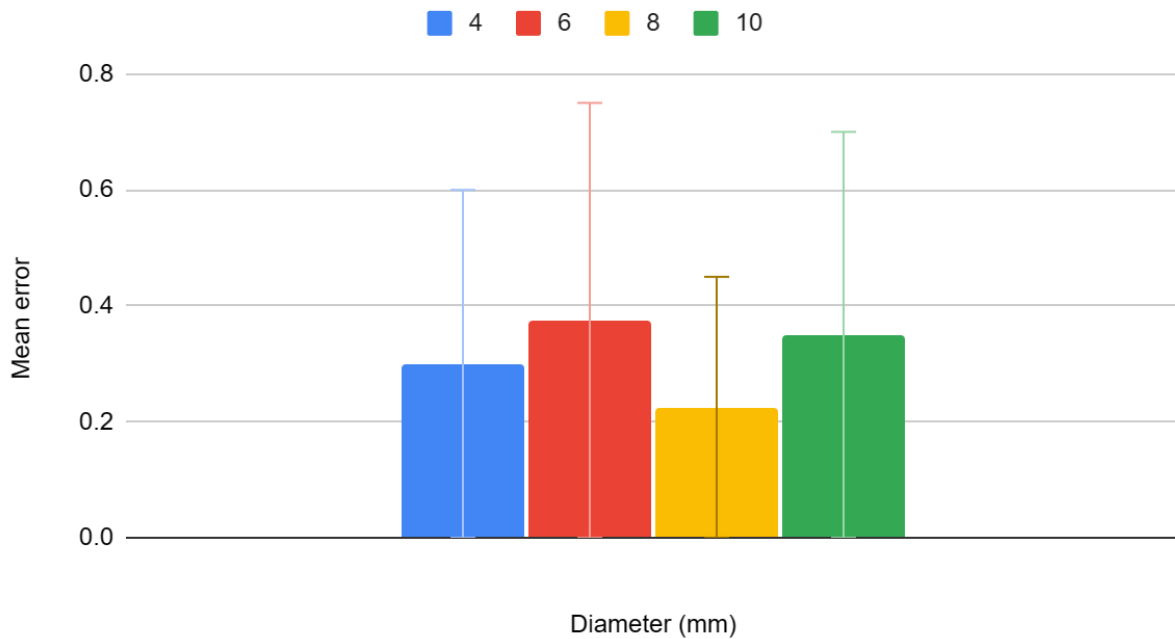


Figure 7. Represents the relationship between the average error and diameter of the hole in the laser cut machine.

Figure 7 shows that higher diameters will result in higher deviations, as there is a positive correlation.

Since tolerances were so small for laser cutting, no tolerances were taken into account when producing the final casing, thus no constraints were expected.

Reflection on initial designs, prototyping and physical testing

Several design ideas were explored before reaching the final selected design. Questions of whether it is possible and sensible to make the chosen design were taken into considerations, along with different cost factors.

Furthermore, as seen in the Design and Professional Skills lectures, Life Cycle Analysis is a method which evaluates the environmental impact associated with a product/activity. It involves looking at:

- **Inventory** (accessibility): limited to the elements provided by the university - 3D printers, laser cutter, and box of components
- **Impact** (efficiency): material comparisons (table below)
- **Improvement**: use as little material as possible (more in the discussion section)

Category 1: The Casing

The first element taken into consideration was the material and design technique used. The table shows the compared cost factors that were taken into consideration.

Cost Factor	Laser cutter: Universal Laser System 60W CO2 Material: Acrylic, also known as polymethyl methacrylate (PMMA)	3D printer: Ultimaker 3 or Ultimaker 3 Extended Material: Polylactic Acid (PLA)
Time	Minutes	Hours
Price of machine	Around £1,300.00 ⁵	Around £2,983.50 ⁶

⁵ LASER ENGRAVER CUTTING MACHINE 700 X 500 60 WATT very good condition with extras. Available at: https://www.ebay.co.uk/itm/256250304937?var=0&mkevt=1&mkcid=1&mkrid=710-53481-19255-0&campid=5338268676&toolid=10044&customid=CjwKCAjw7oeqBhBwEiwALyHLMz1zzdwHsmJkpsAnGuhg2q3ZAMzF3PkDoRcdkHZYL7pG7r-UmMQBAbOcgHcQAvD_BwE

⁶ 3DGBIRE. Ultimaker 3D Printers. Available at: <https://3dgbire.com/collections/ultimaker-3d-printer>

Price of material	100 sheets of acrylic plastic cost £5.40 (according to provided datasheet)	£32.50 for one roll ⁷
Strength ⁸	Flexural strength (<i>how much the material can resist deformation</i> ⁹): 110 MPa Ultimate tensile strength (UTS): 71 MPa	Flexural strength: 80 MPa Ultimate tensile strength (UTS): 50 MPa
Tolerance	Negligible	Needed to be taken into consideration, varies depending on what is printed
Sustainability	Biocompatible, 100% recyclable and non-biodegradable ¹⁰	Recyclable only under very specific conditions ¹¹

The team estimated that the laser printing was significantly more efficient for the casing. The improvement category (justification of chosen designs) of the LCA is discussed in the rest of the report.

Prototype 1

Research on typical peristaltic pumps showed that in many cases, the tube was resting on a U-shaped case. Following such a design for prototype 1 was decided. The straight part of the

⁷ RS. Ultimaker 2.85mm White PLA 3D Printer Filament, 750g. Available at: https://uk.rs-online.com/web/p/3d-printing-materials/1348192?cm_mmc=UK-PLA-DS3A--google--CSS_UK_EN_PMAX_Catch+All--1348192&matchtype=&gclid=CjwKCAjwkY2qBhBDEiwAoQXK5ZkD1SXP55pCsTw5X6z6FxmF3Sp-iWZ90TvEs6bUhudTThfclt1VMxoCVh8QAvD_BwE&qclsrc=aw.ds

⁸ MakeItFrom.com. (2020) PLA vs. PMMA. Available at: <https://www.makeitfrom.com/compare/Polylactic-Acid-PLA-Polylactide/Polymethylmethacrylate-PMMA-Acrylic>

⁹ Science Direct. (2022) *Flexural Strength*. Available at: <https://www.sciencedirect.com/topics/earth-and-planetary-sciences/flexural-strength#:~:text=Flexural%20strength%20or%20bending%20strength,to%20resist%20deformation%20under%20load>

¹⁰ Omenexus. *Plastics and elastomers*. Comprehensive Guide on Polymethyl methacrylate (PMMA or Acrylic). Available at:

<https://omenexus.specialchem.com/selection-guide/polymethyl-methacrylate-pmma-acrylic-plastic#:~:text=Polymethyl%20methacrylate%20is%20highly%20biocompatible,several%20ways%20to%20recycle%20PMMA>.

¹¹ Swiftpak. (2023) Why is PLA not sustainable? Available at: [https://www.swiftpak.co.uk/insights/why-is-pla-not-sustainable#:~:text=While%20Polylactic%20Acid%20\(PLA\)%20is,compostable'%20and%20'recyclable'](https://www.swiftpak.co.uk/insights/why-is-pla-not-sustainable#:~:text=While%20Polylactic%20Acid%20(PLA)%20is,compostable'%20and%20'recyclable').

U-shape had holes which would allow the casing to be screwed on the breadboard and stay intact (view Figure 8).

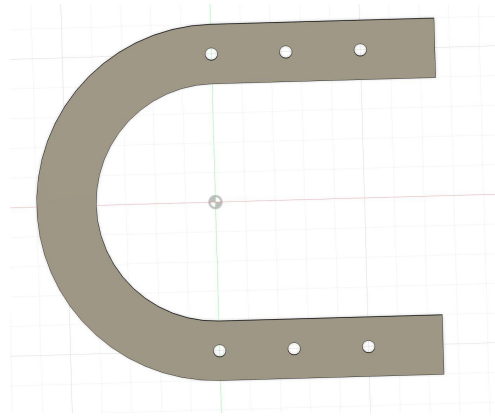


Figure 8. Prototype Case 1, the figure shows one 4 mm layer

To meet the tube's height (6 mm) despite the 4 mm thickness limit of the acrylic sheets, stacking three layers of 4 mm ensured the tube did not fly out when the roller load was applied. A fourth 3 mm thick layer was added which would serve as a cover. The final design is depicted in figure 9.

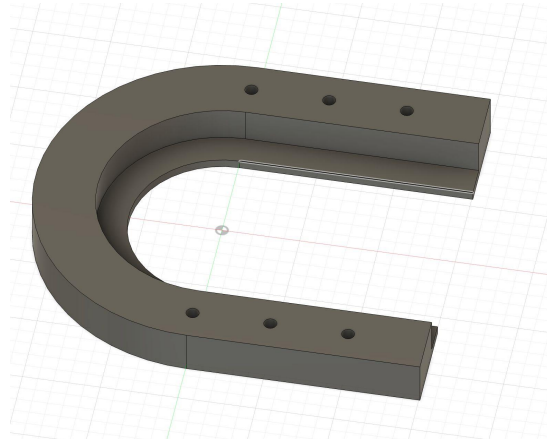


Figure 9. Prototype Case 1 (all layers stacked). Note: the case is upside down, in reality, the cover is at the top not at the base.

Prototype 2

Issues regarding a lack of contact area between the rollers and the tube occurred when using prototype 1, which caused the release of the vacuum and therefore fluid backflow at a certain point. To mitigate this issue, creating a case with a more closed circular contact area was decided (figure 10):

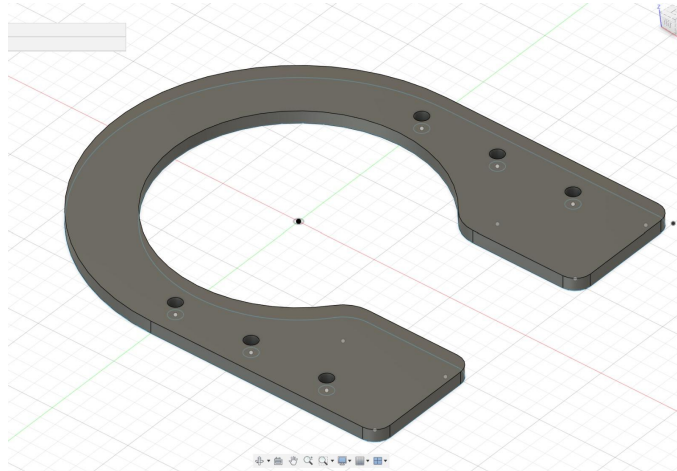


Figure 10. Prototype Case 2 (4 mm thickness). Several layers of this case would be stacked in the final design.

During the testing phase, this design was successful in maintaining a larger contact area with the tube and roller.

Category 2: The Rotator Body

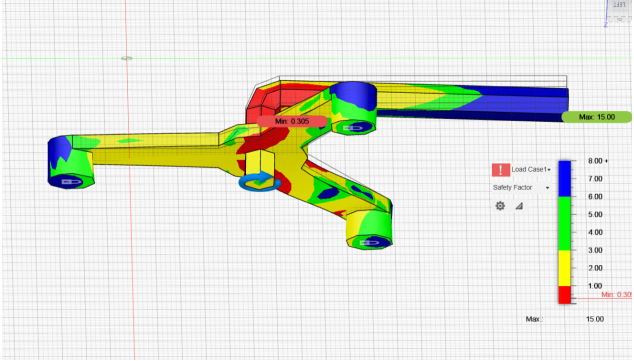
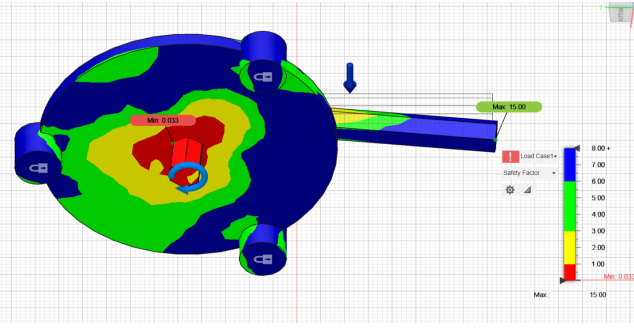
Many designs were considered and analysed using Finite Element Analysis and physical observations when choosing the best design for the rotator body. Static Linear Stress analysis was used with a large force applied (500 Nm moment), with the aim to study a design with a minimum safety factor of 1 or more, ensuring the rotator body can withstand wear and tear after repetitive usage. The table below summarises the 4 considered prototypes in the testing phase.

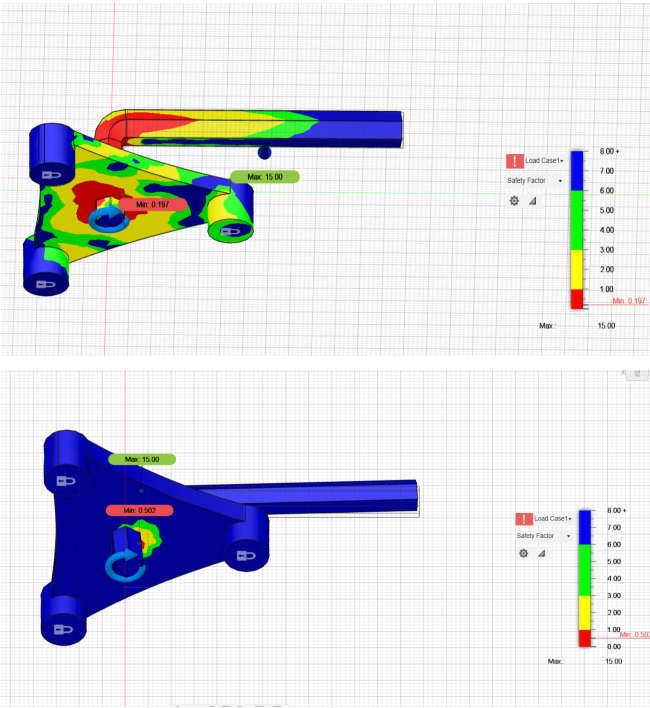
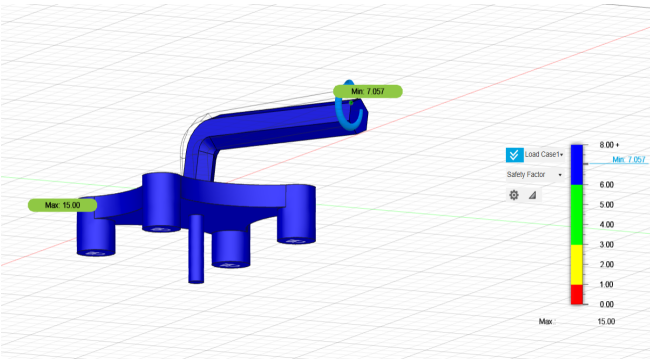
Notes:

- *It was found that the bigger the distance between two rollers, the higher volume of liquid will be contained and the faster the flow will be.¹² It can also be found that using 2 to 3 rollers would increase the flow rate.¹³*
- *All aspects in the testing phase which influenced the final design decision are in bold.*
- *All figures in the table can be found in the appendix (enlarged).*

¹² Premier Control Technologies. (2022) *How To Choose the Correct Peristaltic Pump in Four Steps*. Available at: <https://www.pctflow.com/applications/how-to-choose-the-correct-peristaltic-pump-in-four-steps/#:~:text=The%20number%20of%20rollers%20inside,of%20the%20pump%20head%20housing>.

¹³ Thomas. About Peristaltic Pump Technology. *How Peristaltic Pumps Work?*. Available at: <https://www.thomaspumps.com/en-is/knowledge-hub/peristaltic-pump>

Protot ype Nb.	Nb of wheels	Design shape and FEA Analysis	Safety Factor (needs)	Description/Advantage	Disadvantage	Discussion/ Reflection
1	3, less wheels = increase flow rate (in theory)		Minimum of 0.305 (<1) Very low.	Very thin = minimal material = optimal cost	Stress distribution across area shows rotator would easily break + rotator unable to withstand large and repetitive forces = structural deformation and failure	Design is not reliable. Not printed, not considered.
2	3, less wheels = increase flow rate (in theory)		Minimum of 0.033 (<1) Lowest of all prototypes.	Wheels attached to extrusions at the end of a big circular rotator body + hexagon hole for the Allen key in the middle. In theory, bigger body = better strength	Stress distribution across the area shows the middle part is brittle and easily breakable = results in structural deformation and failure. Too much material used = higher cost and production time.	Design was printed to test the effectiveness of 3 wheels. Tests showed that 3 wheels were inefficient as vacuum was released at a certain point, causing fluid backflow. Also, a big circular body = a lot of friction = inefficient.

3	3, less wheels = increase flow rate (in theory)		For thickness of 3 mm, min. safety factor is 0.197 (<1) For thickness of 5 mm, min. safety factor is 0.502 (<1)	More material than prototype 1, but less than 2 = middle ground for cost in terms of both price and time = optimal amount of material	The 3 wheel design caused a release in vacuum in the tube due to the small contact area between tube and roller, leading to fluid backflow = not functioning	Increasing the thickness of the body also increases the safety factor. This design was not printed because it had 3 wheels, but was designed with FEA to study a balanced amount of material. Design was not built.
4	4, solves the problem of vacuum release and fluid backflow		Safety factor of 7.037 (>1)	Four wheels ensure the tube is in contact with two consecutive wheels at all times. Centre of the body is divided in two: the upper half connects to the Allen Key, and the lower half has a shaft which would be inserted in the breadboard. The shaft's diameter is slightly smaller than that of the hole in the breadboard to allow rotation.	Allen key was not holding onto the body properly as the hexagon hole was not deep enough Bottom shaft was too sensitive and broke = waste of material	Design was built and tested with case 2. Design functioned and pumped water when powered with hand (not Allen Key) Material amount + production time are efficient

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Category 3: The Cap Connecting the Tube to the Liquid-carrying Bags.

The main aspect that needed to be taken into consideration was whether the cap would be connected externally or internally to the bag opening and tube. To answer this question, the team discussed that connecting the tubes on the inside would add material to the diameters of the components, which would restrict the flow of fluid, therefore reducing the flow rate.

The shape was chosen to be simple to minimise costs (figure 16). A general tolerance of 0.5 mm was taken into account when 3D printing this component.

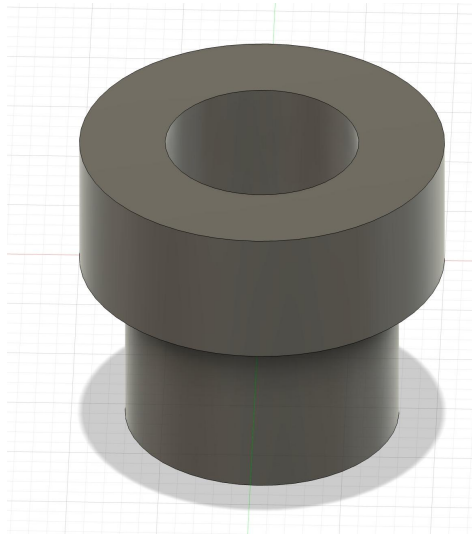


Figure 16. Cap connecting bags and tube

This design functioned well.

Peristaltic pump FEA modelling and animation

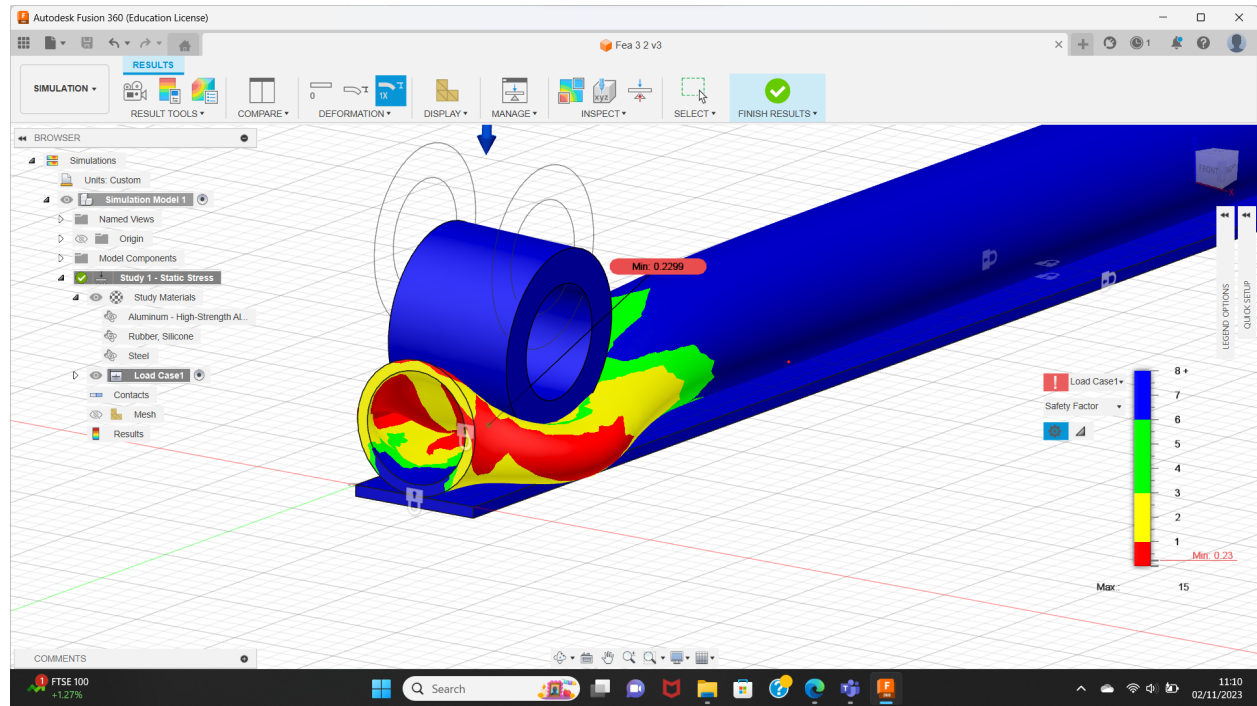


Figure 17. FEA of the roller on the tube

A force of 50N was applied to the stimulation of the bearing and tube to understand how much force was needed to compress the tube. It is clear that 50N is enough for this to compress, therefore there will be a react force similar to this value on the spinner. A proportional value can be applied on the spinners to stimulate a real stimulation and its durability.

The bearing material was aluminium and the tube was rubber silicon since it had similar properties to the silicon tube and the plastic bearing. The weight of the bearing was not added in the stimulation as it acted perpendicular to the reaction force in our prototype.

The animation for the final design can be found in the submitted ZIP file, names “animation.mp4”.

Selected Design Idea For The Pump

Note: TAP drawings are available for every component mentioned in this section in the appendix.

Final Prototype Choices

- For the cap connecting the tube and bag, prototype 1 was chosen, as it proved to function properly with no fluid leakage.

- For the casing, 7 layers of prototype 2 were used, creating a 28 mm thick case. In addition, 1 layer of prototype 1 was used as a cover to prevent the tube from flying out of the case when the load of the rollers is applied.
- For the rotating body, prototype 4 was chosen but changes were made to improve it.

Discussion

For the final design, 8 rollers were used instead of 4, because having 2 superposed rollers in 1 extrusion ensured that the tube would be in contact with the rollers even if it slightly moved up or down. This keeps the tube compressed at all times (cover from the top + superposed wheels). For this, the length of the extrusions holding the wheels increased compared to the initial prototype 4.

Furthermore, prototype 3 of the rotating body compared two thicknesses, showing that a thicker body had a higher safety factor and could withstand higher forces. Thus, the final design included the shape of prototype 4 (which meets the safety factor >1), but with a thicker central body.

Finally, as the new case (prototype 2) was more enclosed, the middle shaft holding the rotating body into the breadboard from prototype 4 was removed. It was deemed to be too sensitive and thus a waste of material. Instead, the full thickness was used for the hole for the Allen Key.

Minimising friction was attempted by limiting the contact areas: only the bottom surface of the wheels touched the breadboard, and only the wheels touched the casing. This was done to prevent heating and wearing.

Figure 18 and 19 show the final assembled pump, and a TAP drawing can be found in the appendix.



Figure 18. Components that will need to be assembled

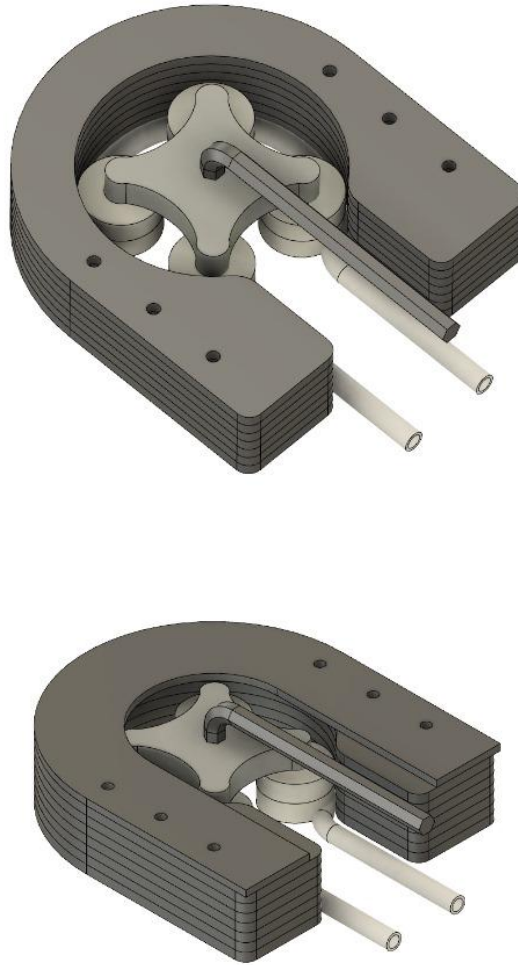


Figure 19. Final design without cover (top) vs with cover (bottom)

Finally, several improvements could be made to the final design to increase its efficiency, examples are:

- Adding gears to allow faster and easier rotations
- Fixing the rotator body in the casing in some way to make it more intact
- Minimising friction by lifting the base of the wheel from the breadboard or by adding a wheel in the centre of the body.

However, they would all require an increase in material used, production time, or cost.

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https://www.ebay.co.uk/itm/256250304937?var=0&mkevt=1&mkcid=1&mkrid=710-53481-19255-0&campid=5338268676&toolid=10044&customid=CjwKCAjw7oeqBhBwEiwALyHLMz1zzdwHsmJkpsAnGuhg2q3ZAMzF3PkDoRcdkHZYL7pG7r-UmMQBABoCGhcQAvD_BwE

3DGBIRE. Ultimaker 3D Printers. Available at:
<https://3dgbire.com/collections/ultimaker-3d-printer>

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https://uk.rs-online.com/web/p/3d-printing-materials/1348192?cm_mmc=UK-PLA-DS3A--googl-e--CSS_UK_EN_PMAX_Catch+All-- --1348192&matchtype=&gclid=CjwKCAjwkY2qBhBDEiWAOQXK5ZkD1SXP55pCsTw5X6z6Fxm3Sp-iWZ90TvEs6bUhudTThfclt1VMxoCVh8QAvD_BwE&gclidsrc=aw.ds

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[https://www.swiftpak.co.uk/insights/why-is-pla-not-sustainable#:~:text=While%20Polylactic%20Acid%20\(PLA\)%20is,compostable'%20and%20'recyclable'.](https://www.swiftpak.co.uk/insights/why-is-pla-not-sustainable#:~:text=While%20Polylactic%20Acid%20(PLA)%20is,compostable'%20and%20'recyclable'.)

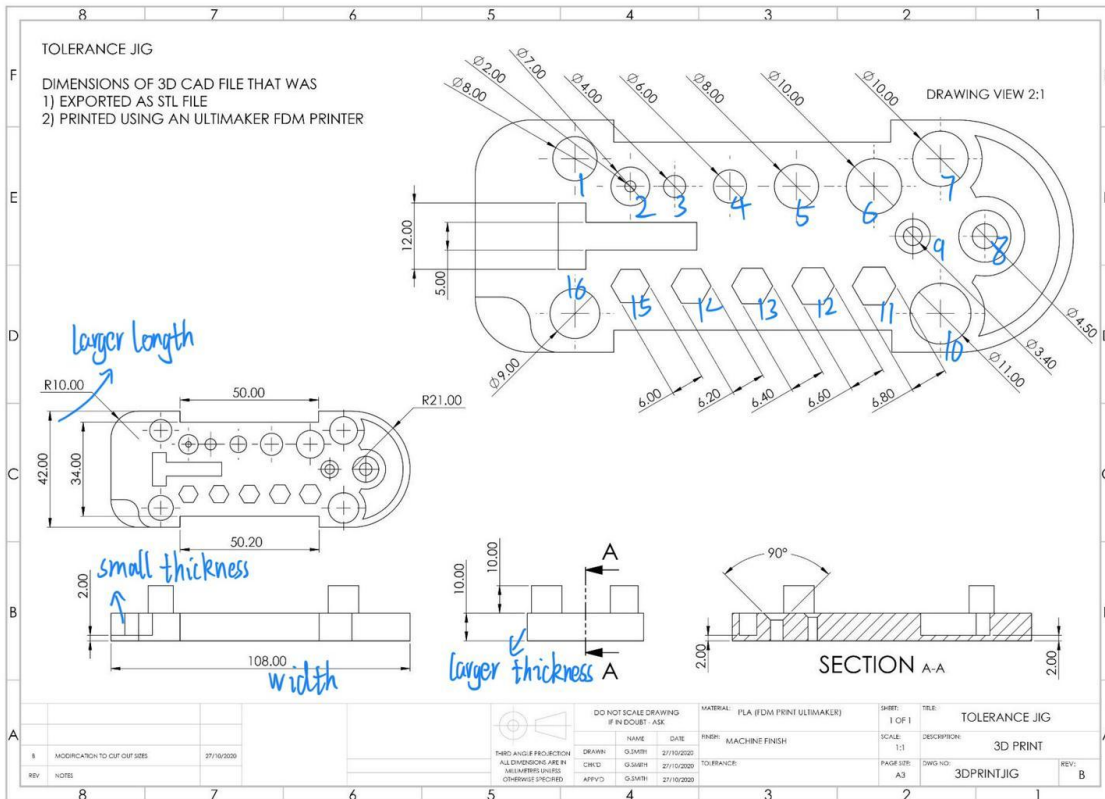
Premier Control Technologies. (2022) *How To Choose the Correct Peristaltic Pump in Four Steps*. Available at:
<https://www.pctflow.com/applications/how-to-choose-the-correct-peristaltic-pump-in-four-steps/#:~:text=The%20number%20of%20rollers%20inside,of%20the%20pump%20head%20housing.>

Thomas. About Peristaltic Pump Technology. *How Peristaltic Pumps Work?*. Available at:
<https://www.thomaspumps.com/en-is/knowledge-hub/peristaltic-pump>

Appendix

1. TOLERANCES MEASUREMENTS AID

Naming and numbering of the elements in the measured shapes (for tolerance analysis):



5.2.1 FDM 3D Print

5.2 Technical drawing for tolerance test pieces

2. BILLS OF MATERIALS

Table of errors for the 3D print cut-out

SHAPE	Operator	Extrusion 1 error	Extrusion 7 error	Extrusion 10 error	Extrusion 16 error	Hexagon 11 error	Hexagon 12 error	Hexagon 13 error	Hexagon 14 error
1	Dong Li	-0.1	-0.2	0	0	-0.3	-0.4	-0.4	-0.3
	Ummu	-0.3	-0.4	-0.2	-0.3	-0.1	-0.4	-0.1	-0.1
	Andrew	-0.6	-0.8	-0.7	-0.6	-0.5	-0.3	-0.3	-0.3
	Lynn	0	0	0	0	-0.3	-0.1	0.4	0
2	Dong Li	-0.1	-0.1	-0.4	-0.5	0.2	0.5	0.6	0.3
	Ummu	-0.3	-0.4	-0.3	-0.4	-0.1	-0.2	-0.4	-0.4

	Andrew	-0.1	-0.2	-0.5	-0.3	0.3	-0.2	0.1	0.3
	Lynn	0	0	0	0	0.2	0.4	0.6	0.8
3	Dong Li	-0.5	-0.5	-0.2	-0.5	0.3	0.4	0.5	0.4
	Ummu	-0.4	-0.4	0	0	-0.2	-0.2	-0.6	-0.2
	Andrew	-0.5	-0.7	-0.1	-0.2	0	-0.4	-0.4	-0.2
	Lynn	-0.4	-0.4	0	-0.1	0.3	0.3	0.1	0.3
4	Dong Li	-0.5	-0.4	-0.2	-0.1	0	-0.4	-0.5	-0.5
	Ummu	-0.3	-0.6	-0.4	-0.4	-0.3	-0.3	-0.4	-0.1
	Andrew	-0.2	-0.4	-0.1	0.1	-0.4	-0.3	-0.2	-0.1
	Lynn	0	-0.2	0	0	0.2	0.4	0.1	-0.1
Actual value		8	10	11	9	6.8	6.6	6.4	6.2
Mean error		-0.26875	-0.35625	-0.19375	-0.20625	-0.04375	-0.075	-0.05625	-0.0125
Standard Deviation		0.20238 16527	0.23084 98791	0.21437 8948	0.22351 36088	0.27072 4337	0.343511 2807	0.40819 72562	0.34423 82896

SHAPE	Operator	Hole 2 error	Hole 3 error	Hole 4 error	Hole 5 error	Hole 6 error
1	Dong Li	-0.5	-0.9	-0.5	-0.6	-0.8
	Ummu	-0.4	-0.6	-0.4	-0.3	-1
	Andrew	-0.7	-0.8	-0.6	-0.9	-1
	Lynn	-0.4	-0.8	-0.5	-0.2	-0.5
2	Dong Li	0.1	0	0.2	0.6	0.5
	Ummu	-0.3	-0.6	-1	-0.5	-0.5
	Andrew	0.1	0.1	0.1	0.1	0.1
	Lynn	0	0	0	1	0.5
3	Dong Li	0.1	0	-0.1	-0.1	0.1
	Ummu	-0.5	-0.5	-0.5	-0.4	-0.4
	Andrew	-0.5	-0.5	-0.5	-0.5	-0.5
	Lynn	0	-0.6	-0.5	-0.1	-0.5
4	Dong Li	-0.6	-0.8	-0.8	-0.2	-0.8
	Ummu	-0.4	-0.6	-0.6	-0.3	-0.6
	Andrew	0.6	0.1	0	0.2	0.3
	Lynn	0	-0.8	0	-1	0

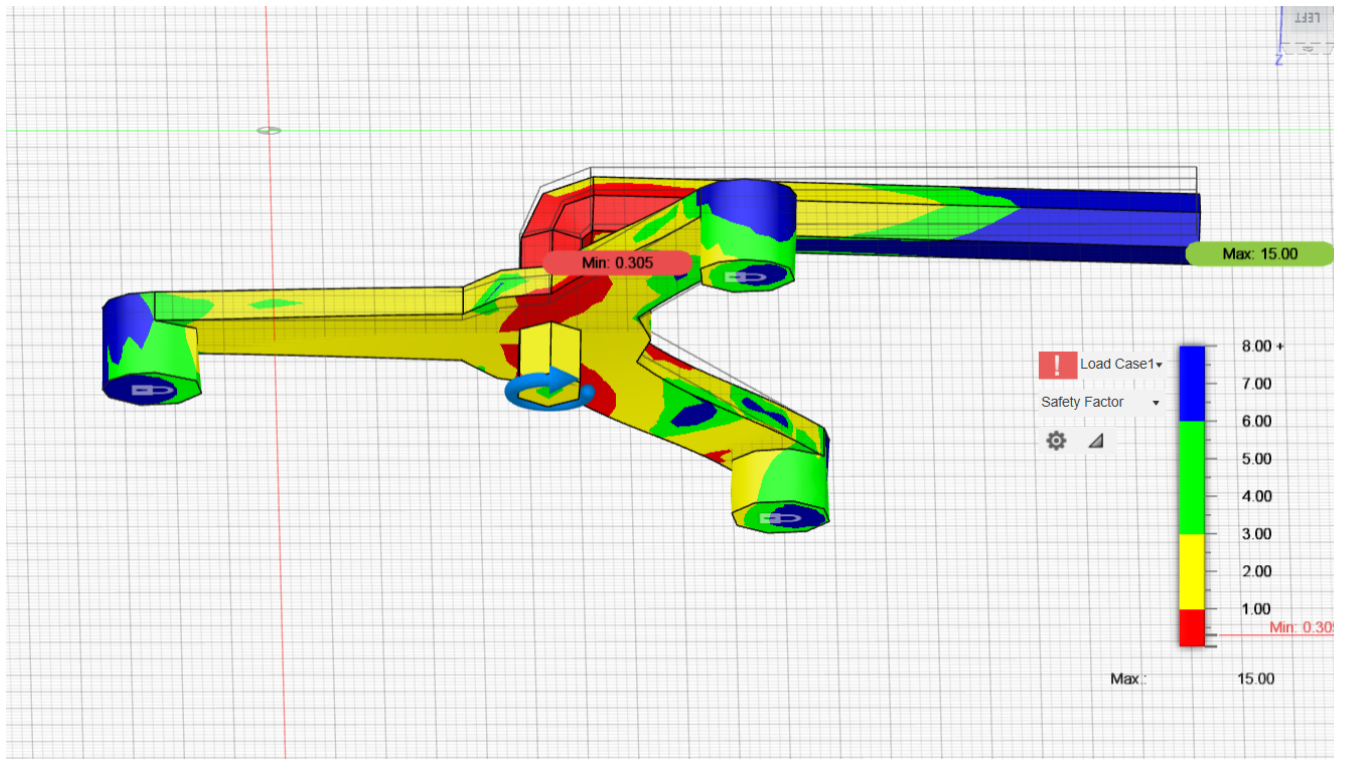
Actual value		7	4	6	8	10
Mean error		-0.2125	-0.45625	-0.35625	-0.2	-0.31875
Standard Deviation		0.35	0.36509 13128	0.34635 00156	0.50859 28299	0.50228 64389

Table of errors for laser cutter

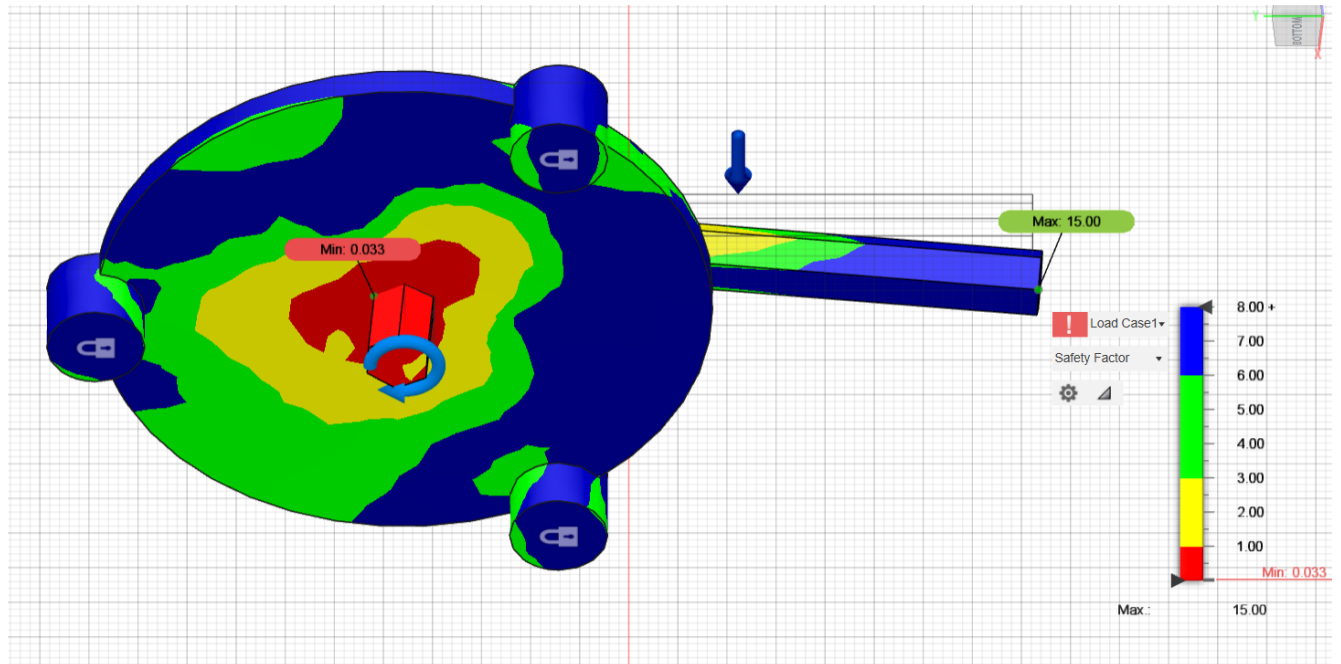
Hole number	2	3	4	5
Diameter(mm)	4	6	8	10
Mean error	0.3	0.375	0.225	0.35
SD	0.08	0.1	0.15	0.06

3. FIGURES FROM TABLE OF PROTOTYPES

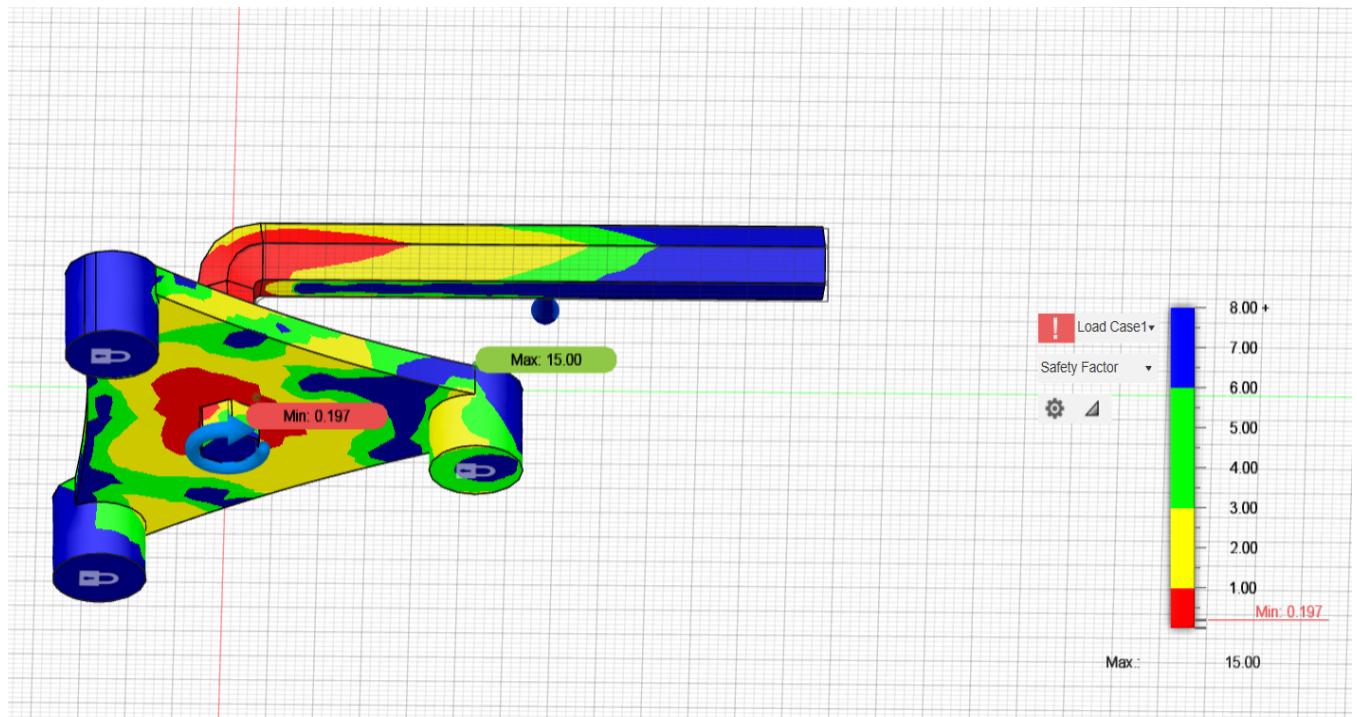
Prototype 1:



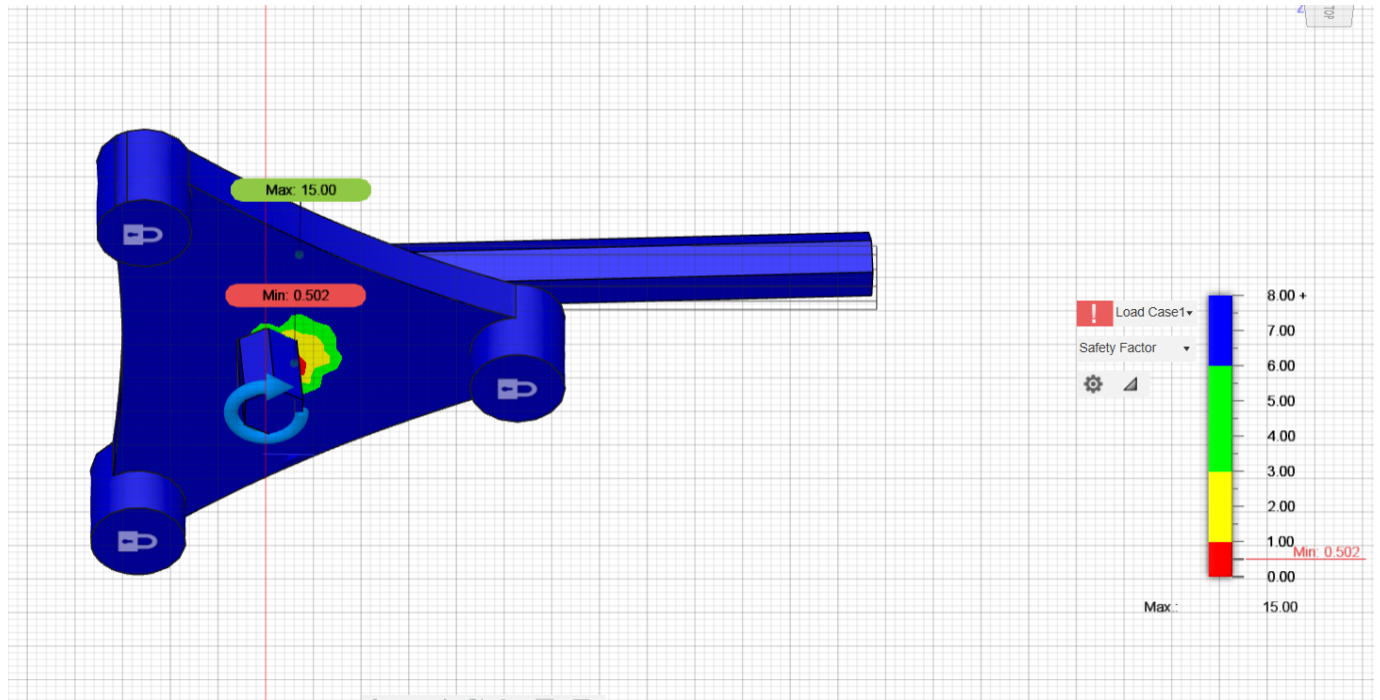
Prototype 2:



Prototype 3:



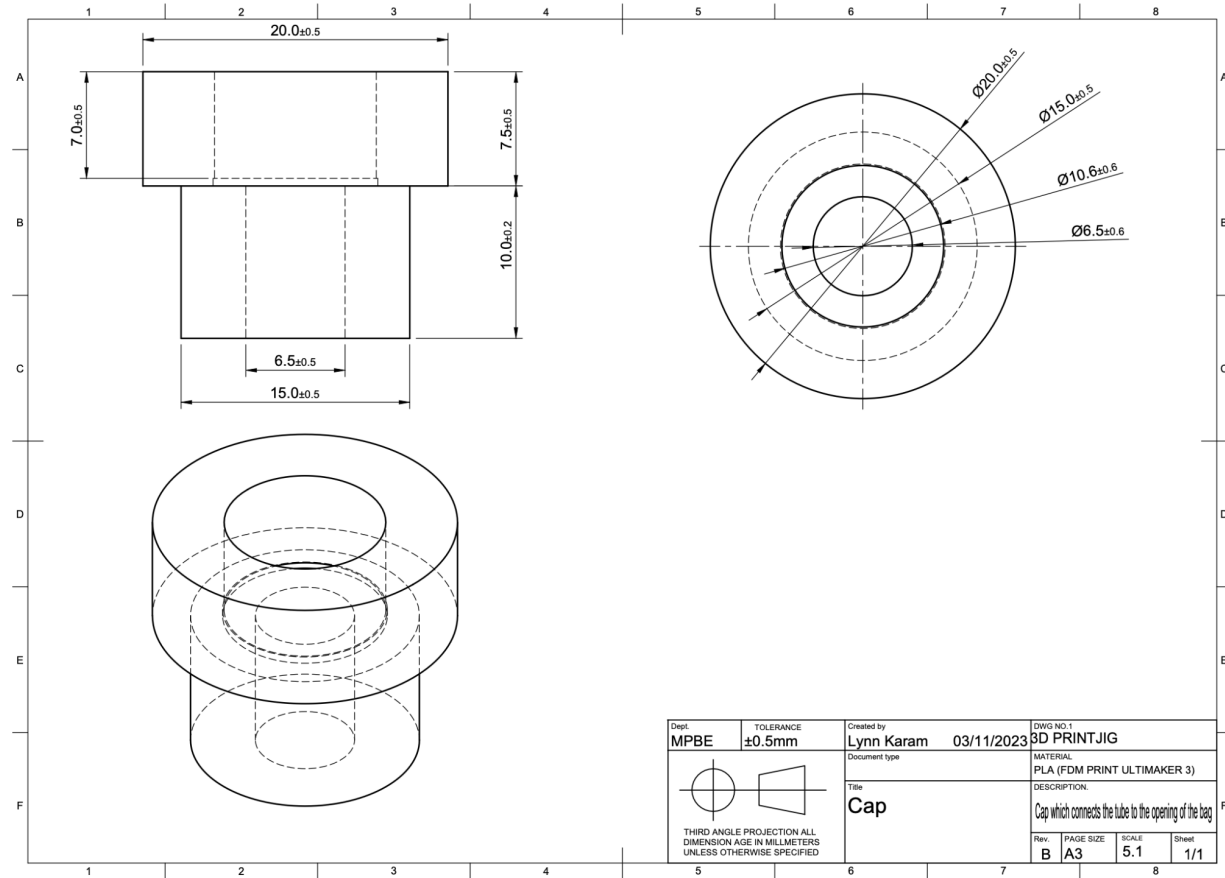
3 mm thickness



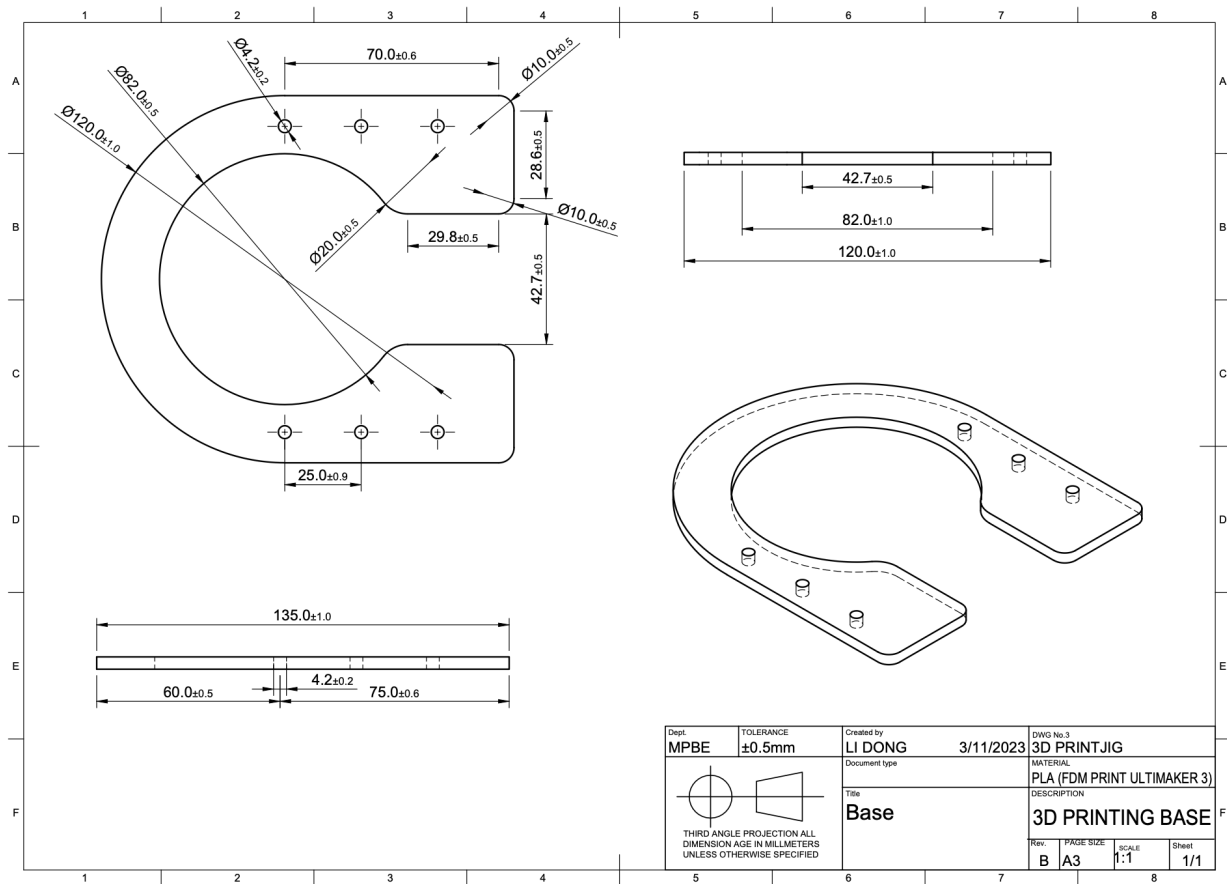
5 mm thickness

4. TAP DRAWINGS FOR FINAL DESIGN COMPONENTS

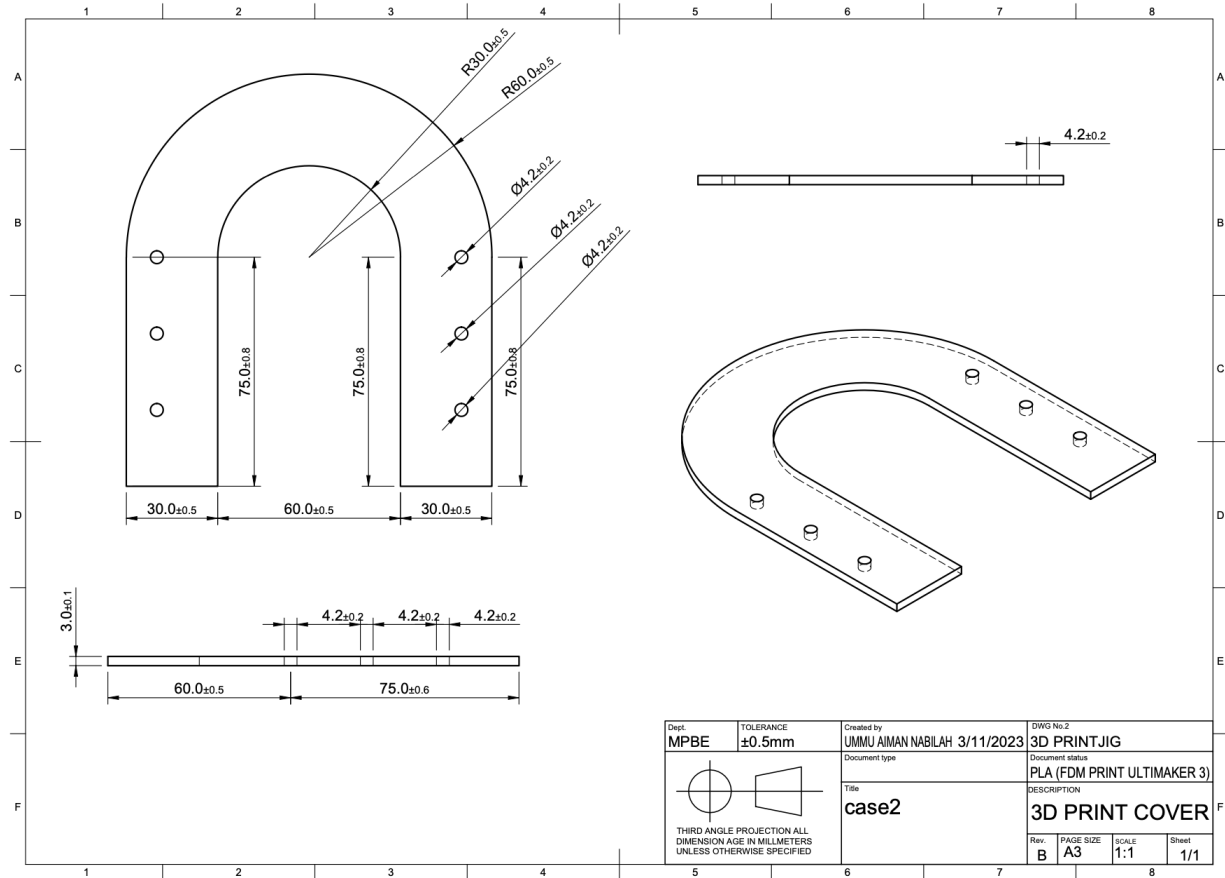
For the cap:



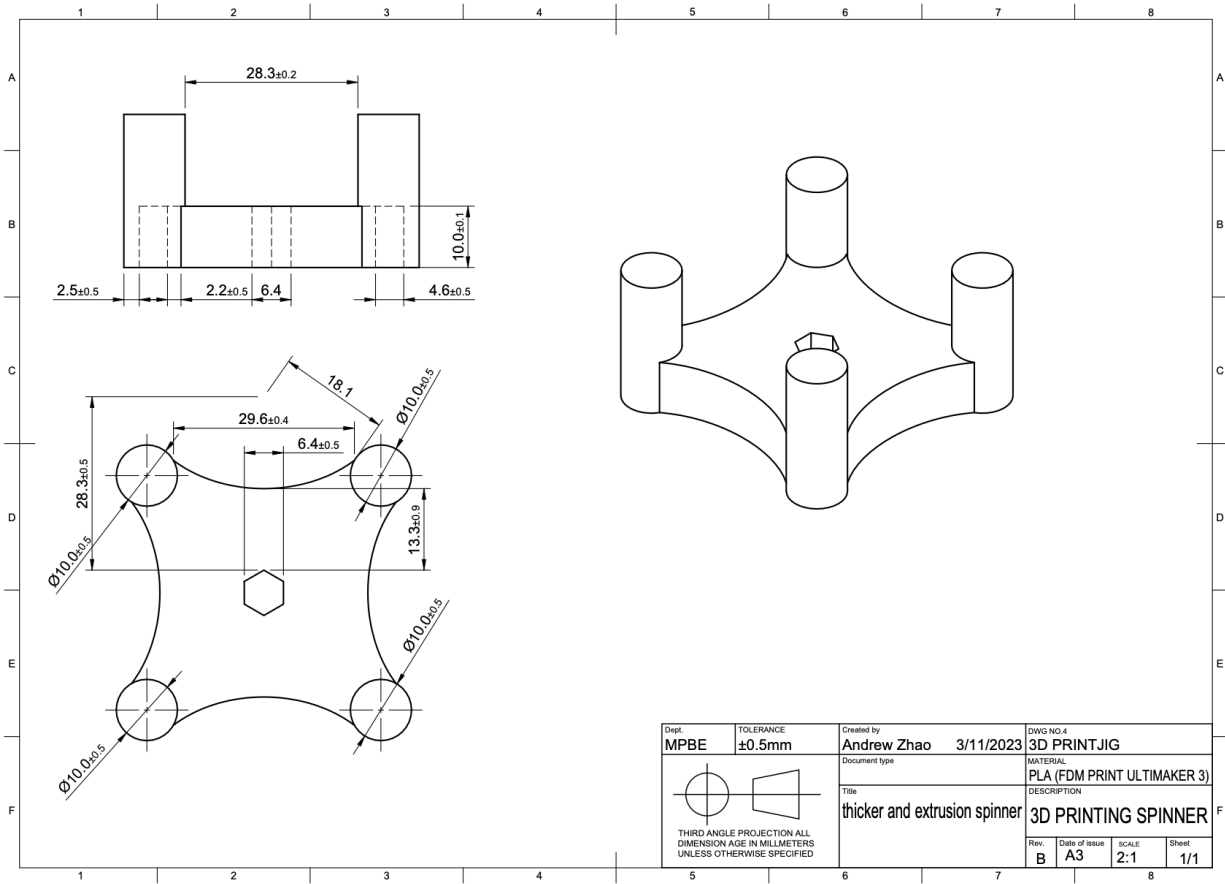
For the base of the casing:



For the cover of the casing:



For the rotator body:



5. TAP DRAWING FOR ASSEMBLED PUMP

